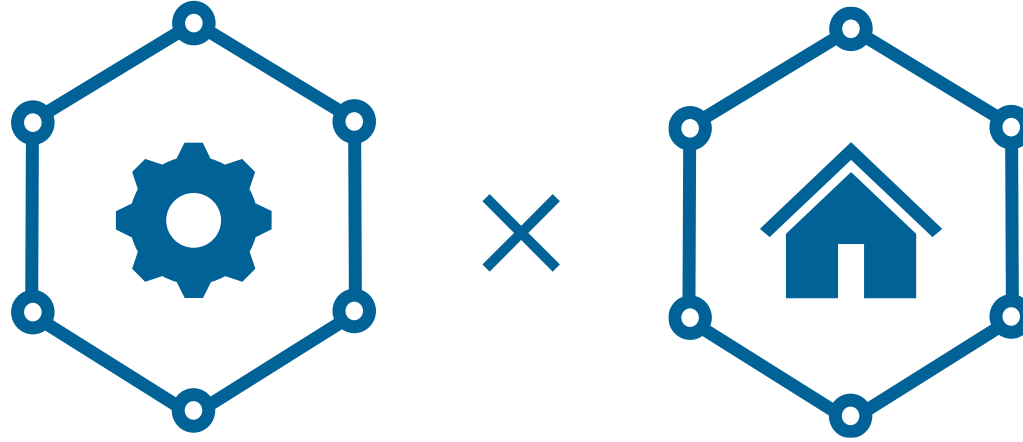
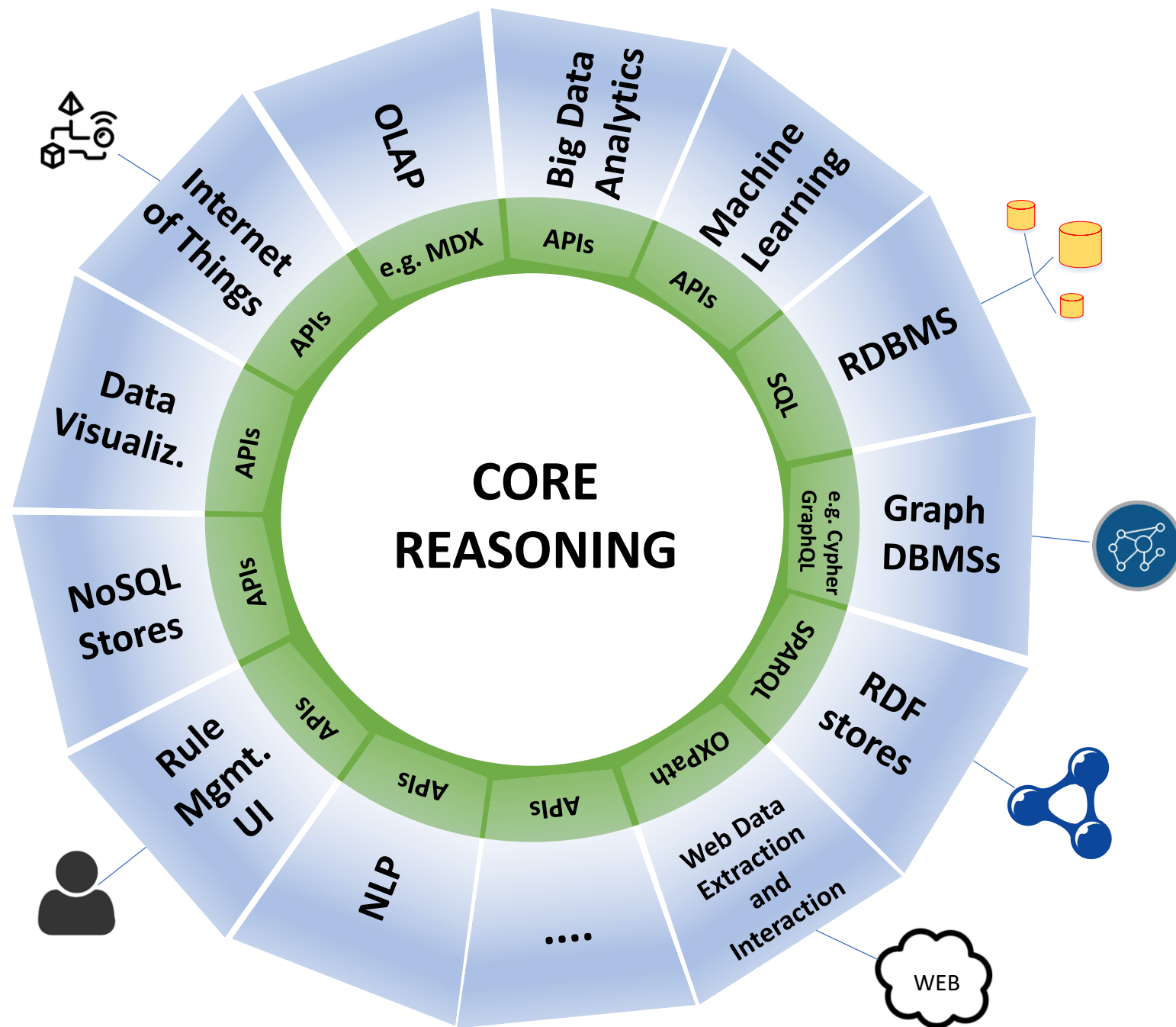
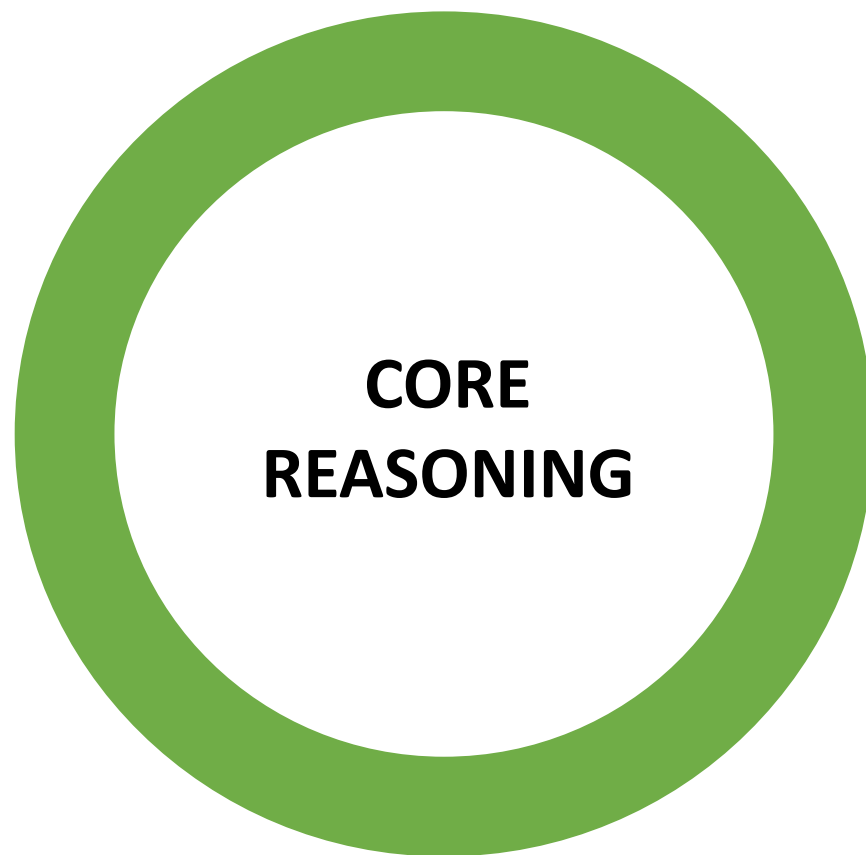




Knowledge Graph Management Systems Reasoning

Emanuel Sallinger







Reasoning Requirements

1. **Recursive** Reasoning: *full recursion over graphs*
2. **Ontological** Reasoning: *object creation, ...*



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**CORE
REASONING**



Vadalog

1. **Recursive** Reasoning:
Full support of recursive **Datalog**
2. **Ontological** Reasoning:
Expressive power of **SPARQL** and **OWL 2 QL**
3. **Scalable** Reasoning: **polynomial time**,
sub-fragments that are **fully parallelizable**



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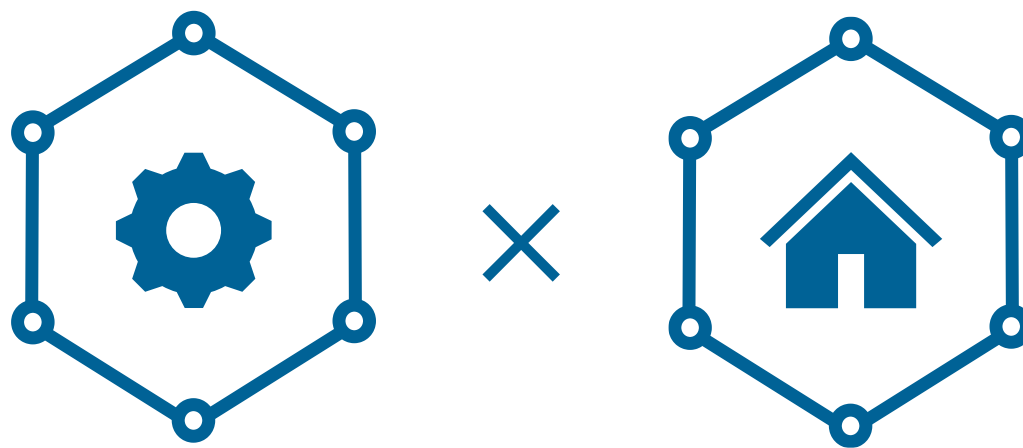
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